

Introduction to Laplace Transform

>> Laplace transform is an integral transform

$$g(x) = \int_a^b f(t) K(x, t) dt$$

o/p i/p integral
kernel

$$x(t) \Rightarrow \underline{x(\omega)} = \int_{-\infty}^{\infty} x(t) \cdot \underbrace{e^{-j\omega t}}_{\text{I.K.}} dt$$

$\omega \rightarrow \text{real variable}$



>> Laplace transform is the general case of Fourier transform

Pierre-Simon Laplace
(1749-1827)

NEW

$$f(t) \Rightarrow \mathcal{L}\{f(t)\} = F(s)$$

$$\underline{F(s)} = \int_{-\infty}^{\infty} f(t) \cdot \underbrace{e^{-st}}_{\text{I.K.}} dt \leftarrow \text{Bilateral LT}$$

o/p i/p damping factor

$$s = \sigma + j\omega$$

↑
Complex variable/freq.
freq. rad/sec.

$$f(t), t \geq 0$$

$$\underline{F(s)} = \int_0^{\infty} f(t) \cdot e^{-st} dt \leftarrow \text{Unilateral LT}$$

$$\rightarrow \text{ILT} : - \int_{\sigma-j\infty}^{\sigma+j\infty} F(s) \cdot e^{st} ds$$

$\times f(t) = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} F(s) \cdot e^{st} ds$

Introduction to Laplace Transform

3.5 lakh views 6 yr ago Laplace Transform | Chapter-6 | ...more



Neso Academy 27.4 lakh



3.1K

81



Share



Remix



Thank

Comments 53



boring neend aajati hae try to make it in interesting way

Relation between Laplace & Fourier Transform

$$\text{LT } s=j\omega \rightarrow \text{FT}$$

$$f(t) \Rightarrow F(\omega)$$

$$f(t) \Rightarrow F(s)$$

$$F(s) = \int_{-\infty}^{\infty} f(t) \cdot e^{-st} dt$$

$$s = \sigma + j\omega$$

$$F(s) = \int_{-\infty}^{\infty} f(t) \cdot e^{-(\sigma+j\omega)t} dt$$

$$F(s) = F\{f(t) \cdot \underbrace{e^{-\sigma t}}_{\sigma=0}\}$$

$$F(s) = F\{f(t) \cdot e^{0t}\}$$

$$F(s) = F\{f(t)\} = F(\omega)$$

224 / Signal & System

5:25



Relation between Laplace Transform &

Next: Relation between Laplace Transform & Fo...

Laplace Transform | Chapter-6 | Signal an... • 1/37



Laplace Transform

>> Laplace transform is an integral transform

$$g(s) = \int_a^b f(t)K(s,t)dt \quad \mathcal{L}\{f(t)\} \Rightarrow \mathcal{X}(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt$$

>> Laplace transform is the general case of Fourier transform

$$f(t) \Rightarrow \mathcal{L}\{f(t)\} = F(s)$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt \quad s = \sigma + j\omega$$

$$s = \sigma + j\omega$$



10:28

223/Signal & System

Introduction to Laplace Transform

Neso Academy

3.5 lakh views · 6 years ago

Relation between Laplace & Fourier Transform

$$f(t) \Rightarrow F(s)$$

$$f(t) \Rightarrow F(s)$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$s = \sigma + j\omega$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-(\sigma + j\omega)t}dt$$

$$F(s) = F(j\omega)$$

$$F(s) = F(j\omega)$$

$$F(s) = F(j\omega)$$

$$F(s) = F(j\omega)$$

224/Signal & System

Relation between Laplace T...

Neso Academy

1.9 lakh views · 6 years ago



Condition for Existence of Laplace Transform

General LT of $f(t) = \mathcal{L}\{f(t)\}$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$$

225/Signal & System

7:25

Condition for Existence of Laplace Transform

Neso Academy

1.4 lakh views · 6 years ago

ROC & its Properties

ROC: It is the range of complex variable 's' in s-plane for which Laplace transform is finite or convergent

$$f(t) \Rightarrow F(s)$$

$$f(t) \Rightarrow F(s)$$

Properties:

1. For right sided signals, ROC is right side of the rightmost pole.



2. For left sided signals, ROC is left side of the leftmost pole.



226/Signal & System

10:56

ROC and its Properties

Neso Academy

2.4 lakh views · 6 years ago

ROC & Stability

Question: Check the stability of LTI system and comment about the extension of $h(t)$

1. $h(t) = e^{2t}u(t)$, ROC: $\sigma > 2$

2. $h(t) = e^{-2t}u(t)$, ROC: $\sigma < -2$

3. $h(t) = e^{2t}u(-t)$, ROC: $\sigma < 2$

4. $h(t) = e^{-2t}u(-t)$, ROC: $\sigma > -2$

5. $h(t) = e^{2t}u(t)$, ROC: $\sigma > 2$

6. $h(t) = e^{-2t}u(t)$, ROC: $\sigma < -2$

7. $h(t) = e^{2t}u(-t)$, ROC: $\sigma < 2$

8. $h(t) = e^{-2t}u(-t)$, ROC: $\sigma > -2$

Homework

5. $h(t) = e^{2t}u(t)$, ROC: $\sigma > 2$

6. $h(t) = e^{-2t}u(t)$, ROC: $\sigma < -2$

7. $h(t) = e^{2t}u(-t)$, ROC: $\sigma < 2$

8. $h(t) = e^{-2t}u(-t)$, ROC: $\sigma > -2$

9. $h(t) = e^{2t}u(t)$, ROC: $\sigma > 2$

10. $h(t) = e^{-2t}u(t)$, ROC: $\sigma < -2$

11. $h(t) = e^{2t}u(-t)$, ROC: $\sigma < 2$

12. $h(t) = e^{-2t}u(-t)$, ROC: $\sigma > -2$

13. $h(t) = e^{2t}u(t)$, ROC: $\sigma > 2$

14. $h(t) = e^{-2t}u(t)$, ROC: $\sigma < -2$

15. $h(t) = e^{2t}u(-t)$, ROC: $\sigma < 2$

227/Signal & System

7:36

ROC and Stability

Neso Academy

1.3 lakh views · 6 years ago

Region of Convergence

Shortcut for Region of

Condition for Existence of Laplace Transform

Bilateral LT of $\underline{f(t)}$ = FT of $\underline{f(t)}e^{-\sigma t}$

$$F(s) = \int_{-\infty}^{+\infty} \underbrace{f(t)e^{-\sigma t}}_{f_1(t)} e^{-j\omega t} dt$$

$$\Rightarrow \underline{F(s)} = \int_{-\infty}^{+\infty} \overbrace{f_1(t)}^{\text{abs. int.}} e^{-j\omega t} dt$$

$$\int_{-\infty}^{+\infty} |f_1(t)| dt < \infty$$

$$\boxed{\int_{-\infty}^{+\infty} |f(t)e^{-\sigma t}| dt < \infty}$$

→ range of $\sigma \Rightarrow \underline{\text{ROC}}$

Example: Find the ROC of signal $\underline{f(t)} = \underline{\exp(2t)u(t)}$

$$f(t) = \boxed{e^{2t} u(t)}$$

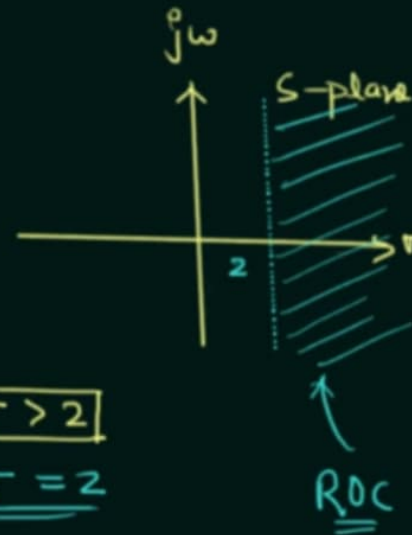
$$\begin{aligned} \int_{-\infty}^{\infty} |f(t)e^{-\sigma t}| dt &= \int_{-\infty}^{\infty} |e^{2t} u(t) e^{-\sigma t}| dt \\ &= \int_0^{\infty} |e^{(2-\sigma)t}| dt \end{aligned}$$

$$\Rightarrow \int_0^{\infty} |e^{\underline{\underline{(2-\sigma)t}}}_{-ve}}| dt < \infty$$

$$2 - \sigma < 0$$

$$-\sigma < -2 \Rightarrow \boxed{\sigma > 2}$$

$$\underline{\underline{\sigma = 2}}$$



ROC: It is the range of complex variable 's' in s-plane for which Laplace transform is finite or convergent

$$\begin{aligned} \checkmark f_1(t) &\Rightarrow F_1(s) \\ \checkmark f_2(t) &\Rightarrow F_2(s) \end{aligned} \left\{ \begin{array}{l} F_1(s) = F_2(s) \\ \sigma > 2 \\ \sigma < -1 \end{array} \right. 4.$$

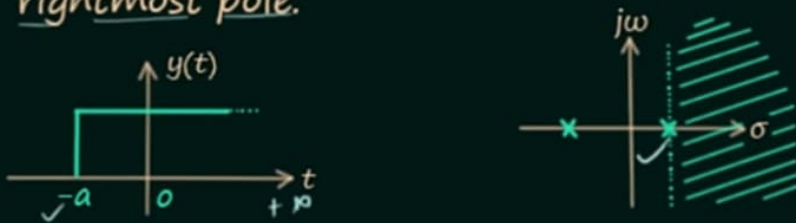
Properties:

1. ROC does not include any poles

$$F(s) = \frac{1}{s+2} \quad s+2=0 \Rightarrow \boxed{s=-2} \quad \text{ROC}$$

$$F(-2) = \frac{1}{-2+2} = \frac{1}{0} \rightarrow \infty$$

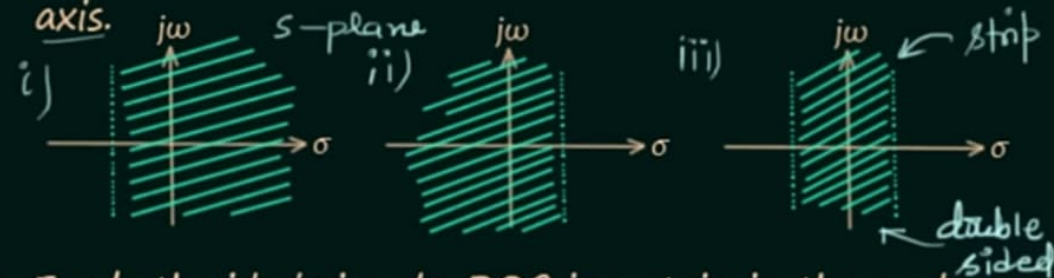
2. For right sided signals, ROC is right side to the rightmost pole.



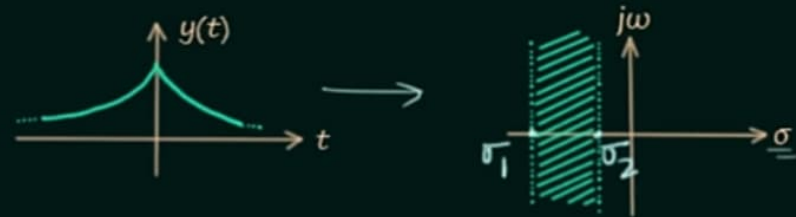
3. For left sided signals, ROC is left side to the leftmost pole.



4. For the absolute integrability of a signal or the stability of a system, ROC should include imaginary axis.



5. For both sided signals, ROC is a strip in the s-plane.



6. For finite duration signals, ROC is the entire s-plane excluding $s = 0$ &/or $+\infty$ &/or $-\infty$.

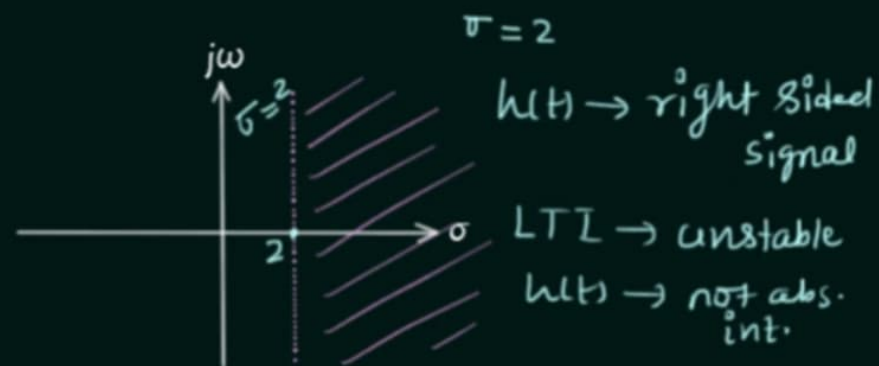


$$f(t) \begin{cases} \xleftrightarrow{x(t)} X(s), \text{ROC} \\ \xleftrightarrow{h(t)} H(s), \text{ROC} \end{cases}$$

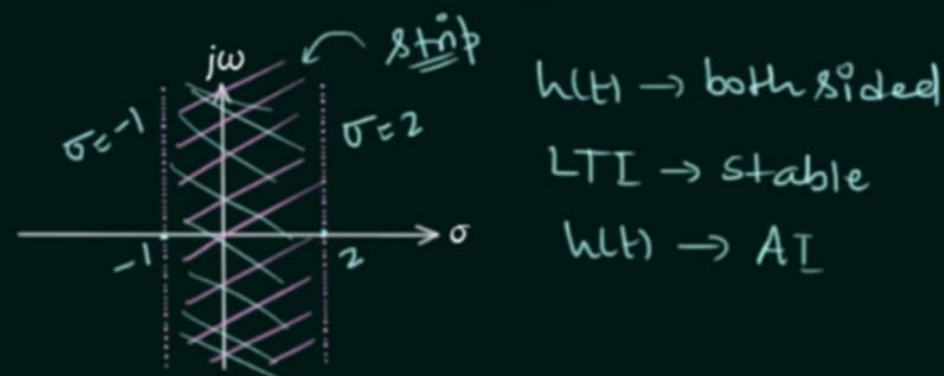
ROC and Stability

Question: Check the stability of LTI system and comment about the extension of $h(t)$

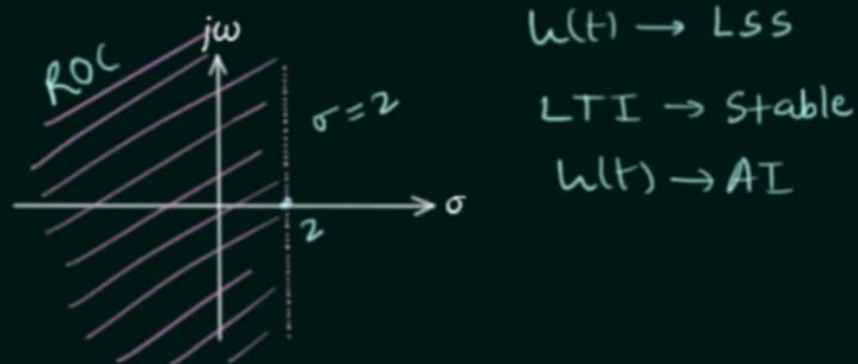
1. $\underline{h(t)} \Rightarrow \underline{H(s)}$, ROC: $\sigma > 2$ 'jw'



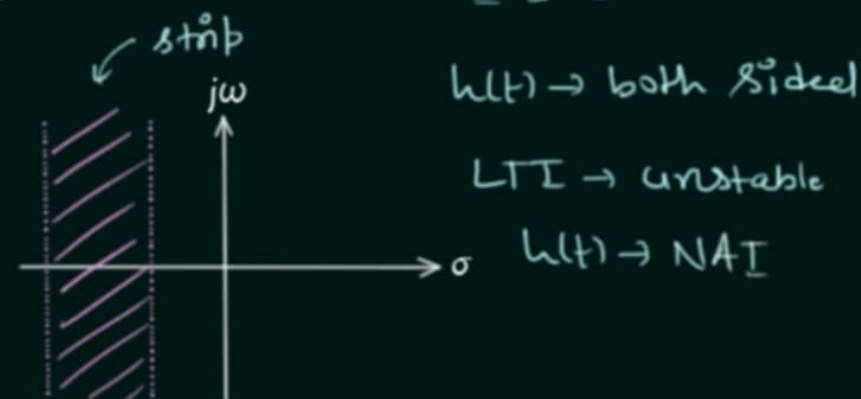
3. $\underline{h(t)} \Rightarrow \underline{H(s)}$, ROC: $-1 < \sigma < 2$



2. $\underline{h(t)} \Rightarrow \underline{H(s)}$, ROC: $\sigma < 2$



4. $\underline{h(t)} \Rightarrow \underline{H(s)}$, ROC: $-3 < \sigma < -2$



Shortcut for ROC

$$\frac{f(t)}{\text{LS}} = \underbrace{e^{-at}}_{\text{LS}} \underbrace{u(-t)}_{\text{Re}}$$

Coeff. of $t = -a$
Re

✓ **Step 1:** Compare σ with the real part of the coefficient of t in power of e

$$\boxed{\sigma < -a} \text{ ROC}$$

✓ **Step 2:** Check if the signal is left sided or right sided and decide < or >

Ex. 1:- $\frac{\text{LS}}{f(t)} = \underbrace{e^{(5+3j)t}}_{\text{LS}} \underbrace{u(-t-5)}_{\text{LSS}}$

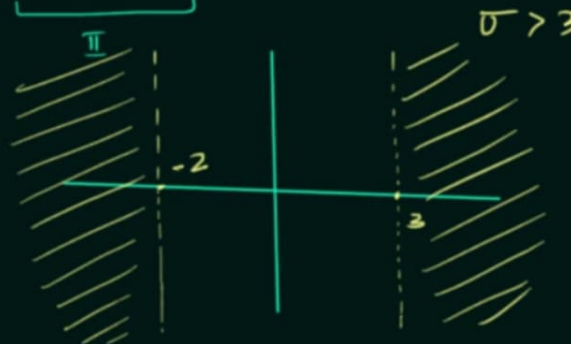
Sol:- $\boxed{\sigma < 5} \text{ ROC}$

H.W. Ex. 3:- $f(t) = -e^{-2t}u(-t) + e^{-3t}u(t)$

Ex. 2:- $f(t) = \underbrace{e^{3t}u(t)}_{\text{I}} + \underbrace{e^{-2t}u(-t)}_{\text{II}}$

No common ROC $\Rightarrow$ No LT

Sol:-
I. $\boxed{\sigma > 3} \text{ ROC 1}$
II. $\boxed{\sigma < -2} \text{ ROC 2}$



Properties of Laplace Transform (Part 1)

1. Linearity: $f_1(t) \Rightarrow F_1(s), \text{ROC} = R_1$
 $f_2(t) \Rightarrow F_2(s), \text{ROC} = R_2$
 $\alpha f_1(t) \Rightarrow \alpha F_1(s), \text{ROC} = R_1 \checkmark$
 $\beta f_2(t) \Rightarrow \beta F_2(s), \text{ROC} = R_2 \checkmark$
 $\alpha f_1(t) + \beta f_2(t) \Rightarrow \alpha F_1(s) + \beta F_2(s),$
 $\underbrace{f(t)}_{\Rightarrow F(s)} \Rightarrow \underbrace{F(s)}_{\text{ROC} = R_1 \cap R_2}$

proof: -

$$F(s) = \int_{-\infty}^{\infty} f(t) \cdot e^{-st} dt$$

$$F(s) = \int_{-\infty}^{\infty} [\alpha f_1(t) + \beta f_2(t)] \cdot e^{-st} dt$$

$$\underline{F(s)} = \alpha \underbrace{\int_{-\infty}^{\infty} f_1(t) \cdot e^{-st} dt}_{F_1(s)} + \beta \underbrace{\int_{-\infty}^{\infty} f_2(t) \cdot e^{-st} dt}_{F_2(s)}$$

2. Conjugation: $f(t) \Rightarrow F(s)$
 $\checkmark \checkmark \quad f^*(t) \Rightarrow F^*(s^*)$

proof: -

$$F(s) = \int_{-\infty}^{\infty} f(t) \cdot e^{-st} dt$$

$$F^*(s) = \int_{-\infty}^{\infty} f^*(t) \cdot e^{-\overset{\text{by } s}{s^*} t} dt$$

$$s \rightarrow s^*$$

$$F^*(s^*) = \int_{-\infty}^{\infty} f^*(t) \cdot e^{-(s^*)^* t} dt$$

$$\underbrace{F^*(s^*)}_{\text{o/p}} = \int_{-\infty}^{\infty} \underbrace{f^*(t)}_{\text{i/p}} \cdot \underbrace{e^{-st}}_{\text{I k}} dt$$

Laplace Transform | Chapte... · 8/37

Neso Academy



Laplace Transform Properties **PART 4**

7. Convolution in Time:
 $f(t) \otimes g(t) \iff F(s)G(s), \text{ ROC} = R_1 \cap R_2$
 $f(t) \otimes g(t) \iff F(s)G(s), \text{ ROC} = R_1 \cup R_2$
 $\frac{f(t) \otimes g(t)}{f(t)} \iff \frac{F(s)G(s)}{F(s)}, \text{ ROC} = R_1 \cap R_2$

8. Multiplication in Time:
 $f(t)g(t) \iff \frac{1}{2\pi j} \int_{\gamma} F(s)G(s)ds, \text{ ROC} = R_1 \cap R_2$

9. Differentiation in Time:
 $f(t) \iff F(s), \text{ ROC} = R$
 $\frac{d}{dt}f(t) \iff sF(s), \text{ ROC} = R$
 $\frac{d^n}{dt^n}f(t) \iff s^n F(s) - s^{n-1}f(0^-) - s^{n-2}\frac{df(0^-)}{dt} - \dots - s\frac{d^{n-1}f(0^-)}{dt^{n-1}}$

232 / Signal & System **8:31**

Properties of Laplace Transform (Part 4)

Neso Academy

Laplace Transform Properties **PART 5**

10. Integration in Time:
 $f(t) \iff F(s), \text{ ROC} = R$
 $\int_{-\infty}^t f(\tau)d\tau \iff \frac{F(s)}{s}, \text{ ROC} = R \cap \text{Re}(s) < 0$
 $\int_t^{\infty} f(\tau)d\tau \iff \frac{F(s)}{s}, \int_{-\infty}^0 f(\tau)d\tau$

233 / Signal & System **9:30**

Properties of Laplace Transform (Part 5)

Neso Academy

81K views · 6 years ago

Laplace Transform Properties **PART 6**

11. Differentiation in Frequency:
 $f(t) \iff F(s), \text{ ROC} = R$
 $t^n f(t) \iff (-1)^n \frac{d^n F(s)}{ds^n}, \text{ ROC} = R$
 $\int_{-\infty}^{\infty} f(t) e^{-st} dt = F(s)$
 $\frac{d}{ds} F(s) = \int_{-\infty}^{\infty} f(t) (-t) e^{-st} dt$
 $\frac{d^n F(s)}{ds^n} = \int_{-\infty}^{\infty} f(t) (-t)^n e^{-st} dt$

12. Integration in Frequency:
 $f(t) \iff F(s)$
 $\frac{f(t)}{t} \iff \int_s^{\infty} F(u)du$
 $\int_{-\infty}^{\infty} f(t) e^{-st} dt = F(s)$
 $\int_{-\infty}^{\infty} f(t) e^{-st} dt = F(s)$

234 / Signal & System **8:50**

Properties of Laplace Transform (Part 6)

Neso Academy

76K views · 6 years ago

Initial Value Theorem

13. Initial Value Theorem:
 $f(t) \iff F(s)$
 $f(0^+) = \lim_{s \rightarrow \infty} sF(s)$

Conditions: 1. Applicable only when $f(t) = 0, t < 0$
2. $f(t)$ must not contain any impulse or higher order singularities at $t = 0$

235 / Signal & System **12:47**

Initial Value Theorem

Neso Academy

1.3 lakh views · 6 years ago

Final Value Theorem

14. Final Value Theorem:
 $f(t) \iff F(s)$
 $\lim_{t \rightarrow \infty} f(t) = f(\infty) = \lim_{s \rightarrow 0} sF(s)$

Final Value Theorem

Neso Academy



Laplace Transform Properties

PART 1

1. Linearity: $f_1(t) \Rightarrow F_1(s), \text{Re } s = R_1$
 $f_2(t) \Rightarrow F_2(s), \text{Re } s = R_2$
 $a f_1(t) + b f_2(t) \Rightarrow a F_1(s) + b F_2(s), \text{Re } s = R$
 $R = \max(R_1, R_2)$

2. Conjugation:

$$f(t) \Rightarrow F(s)$$

$$f^*(t) \Rightarrow F^*(s^*)$$

3. Time Shifting:

$$f(t-t_0) \Rightarrow e^{-st_0} F(s)$$

$$F(s) \Rightarrow f(t)$$

8:49

229 Signal & System

Properties of Laplace Transform (Part 1)

Neso Academy

1.7 lakh views · 6 years ago

Laplace Transform Properties

PART 2

3. Time Reversal:

$$f(t) \Rightarrow F(s), \text{Re } s = R$$

$$f(-t) \Rightarrow F(-s), \text{Re } s = -R$$

4. Time Scaling:

$$f(at) \Rightarrow \frac{1}{|a|} F\left(\frac{s}{a}\right), \text{Re } s = |a|R$$

$$f\left(\frac{t}{a}\right) \Rightarrow |a| F(as), \text{Re } s = R$$

9:54

230 Signal & System

Properties of Laplace Transform (Part 2)

Neso Academy

1 lakh views · 6 years ago

Laplace Transform Properties

PART 3

5. Time Shifting:

$$f(t) \Rightarrow F(s), \text{Re } s = R$$

$$f(t+t_0) \Rightarrow e^{st_0} F(s), \text{Re } s = R$$

6. Frequency Shifting:

$$f(t) \Rightarrow F(s), \text{Re } s = R$$

$$e^{j\omega_0 t} f(t) \Rightarrow F(s-j\omega_0), \text{Re } s = R$$

231 Signal & System

Properties of Laplace Transf...

Neso Academy

93K views · 6 years ago



Laplace Transform Properties

PART 4

7. Convolution in Time:

$$f_1(t) \Rightarrow F_1(s), \text{Re } s = R_1$$

$$f_2(t) \Rightarrow F_2(s), \text{Re } s = R_2$$

$$\frac{f_1(t) * f_2(t)}{t} \Rightarrow \frac{F_1(s) F_2(s)}{s}, \text{Re } s = R$$

8. Differentiation in Time:

$$f(t) \Rightarrow F(s), \text{Re } s = R$$

$$\frac{d f(t)}{dt} \Rightarrow s F(s) - f(0^-), \text{Re } s = R$$

$$\frac{d^2 f(t)}{dt^2} \Rightarrow s^2 F(s) - s f(0^-) - f'(0^-), \text{Re } s = R$$

9. Multiplication in Time:

$$f_1(t) f_2(t) \Rightarrow \frac{1}{2\pi j} \int_{R_1-j\infty}^{R_1+j\infty} F_1(s) F_2(s-s_1) ds_1, \text{Re } s = R$$

$$f_1(t) \Rightarrow F_1(s), \text{Re } s = R_1$$

$$f_2(t) \Rightarrow F_2(s), \text{Re } s = R_2$$

8:31

232 Signal & System

Properties of Laplace Transform (Part 4)

Neso Academy

90K views · 6 years ago

Laplace Transform Properties

PART 5

10. Integration in Time:

$$f(t) \Rightarrow F(s), \text{Re } s = R$$

$$f(t) = \int_{-\infty}^t f_1(\tau) d\tau \Rightarrow \frac{F_1(s)}{s}, \text{Re } s = R$$

$$\int_{-\infty}^t f_1(\tau) d\tau \Rightarrow \frac{F_1(s)}{s} + \frac{f_1(0^-)}{s}, \text{Re } s = R$$

9:30

233 Signal & System

Properties of Laplace Transform (Part 5)

Neso Academy

81K views · 6 years ago

Laplace Transform

PART

Properties of Laplace



Laplace Transform Properties **PART 2**

3. Time Reversal:

$$f(t) \Rightarrow F(s), \text{ROC} = R$$

$$f(-t) \Rightarrow F(-s), \text{ROC} = -R$$

4. Time Scaling:

$$f(t) \Rightarrow F(s), \text{ROC} = R$$

$$f(at), a \neq 0 \Rightarrow \frac{1}{|a|} F\left(\frac{s}{a}\right), \text{ROC} = |a|R$$

230 Signal & System **9:54**

Properties of Laplace Transform (Part 2)

Neso Academy

1 lakh views · 6 years ago

Laplace Transform Properties **PART 3**

5. Time Shifting:

$$f(t) \Rightarrow F(s), \text{ROC} = R$$

$$f(t-t_0) \Rightarrow F(s)e^{-st_0}, \text{ROC} = R$$

6. Frequency Shifting:

$$f(t) \Rightarrow F(s), \text{ROC} = R$$

$$e^{j\omega_0 t} f(t) \Rightarrow F(s-j\omega_0), \text{ROC} = R + j\omega_0$$

231 Signal & System **7:01**

Properties of Laplace Transform (Part 3)

Neso Academy

93K views · 6 years ago

Laplace Transform Properties **PART 4**

7. Convolution in Time:

$$f(t) \Rightarrow F(s), \text{ROC} = R_1$$

$$g(t) \Rightarrow G(s), \text{ROC} = R_2$$

$$f(t) * g(t) \Rightarrow F(s)G(s), \text{ROC} = R_1 \cap R_2$$

8. Multiplication in Time:

$$f(t)g(t) \Rightarrow \frac{1}{2\pi j} \int_{\gamma-j\infty}^{\gamma+j\infty} F(s)G(s-s') ds', \text{ROC} = R_1 \cap R_2$$

9. Differentiation in Time:

$$f(t) \Rightarrow F(s), \text{ROC} = R$$

$$\frac{d}{dt} f(t) \Rightarrow sF(s), \text{ROC} = R$$

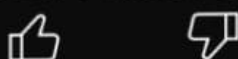
$$t^n f(t) \Rightarrow (-1)^n \frac{d^n}{ds^n} F(s), \text{ROC} = R$$

232 Signal & System

Properties of Laplace Transf...

Neso Academy

90K views · 6 years ago



Laplace Transform Properties **PART 5**

10. Integration in Time:

$$f(t) \Rightarrow F(s), \text{ROC} = R$$

$$f(t) = \int_{-\infty}^t g(\tau) d\tau \Rightarrow \frac{F(s)}{s}, \text{ROC} = R \cap \text{Re}(s) > \sigma_0$$

$$\int_{-\infty}^t f(\tau) d\tau \Rightarrow \frac{F(s)}{s} + \int_{-\infty}^0 f(\tau) d\tau$$

233 Signal & System **9:30**

Properties of Laplace Transform (Part 5)

Neso Academy

81K views · 6 years ago

Laplace Transform Properties **PART 6**

11. Differentiation in Frequency:

$$f(t) \Rightarrow F(s), \text{ROC} = R$$

$$t^n f(t) \Rightarrow (-1)^n \frac{d^n}{ds^n} F(s), \text{ROC} = R$$

12. Integration in Frequency:

$$f(t) \Rightarrow F(s)$$

$$\frac{f(t)}{t} \Rightarrow \int_s^{\infty} F(\sigma) d\sigma$$

Properties of Laplace Transform (Part 6)



2) $f(t) = -e^{-at}u(-t)$
 3) $f(t) = e^{at}u(t)$
 4) $f(t) = -e^{at}u(-t)$

$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st}dt$
 $= 0 + \int_{-\infty}^0 e^{-at}e^{-st}dt$
 $= \int_0^{\infty} e^{-as}e^{-st}dt$

240 Signal & System **9:51**

Signal & System (Exponential Signal),
 Neso Academy
 1.3 lakh views · 6 years ago

**Laplace Transform
(Unit Ramp Signal)**

$f(t) = \begin{cases} 0 & t < 0 \\ t & t \geq 0 \end{cases} = \frac{1}{s}u(t)$

$\rightarrow u(t) \Rightarrow \frac{1}{s}$; ROC : $\text{Re}\{s\} > 0$

$f(t) = \int_{-\infty}^t u(\tau)d\tau \Rightarrow \frac{1}{s} = \frac{1}{s^2}$; ROC : $\text{Re}\{s\} > 0$

241 Signal & System **6:58**

Laplace Transform of Basic
 Signals (Unit Ramp Signal)
 Neso Academy
 73K views · 6 years ago

**Laplace Transform
[Cos $\omega_0 t u(t)$]**

$f(t) = \cos(\omega_0 t)u(t) \Rightarrow F(s) = ?$; ROC = ?

$\cos(\omega_0 t) = \frac{1}{2} [e^{j\omega_0 t} + e^{-j\omega_0 t}]$

$f(t) = \frac{1}{2} [e^{j\omega_0 t} + e^{-j\omega_0 t}] u(t)$

$F(s) = \frac{1}{2} [e^{j\omega_0 t} u(t) + e^{-j\omega_0 t} u(t)]$

242 Signal & System **11:40**

Laplace Transform of Basic
 Signals [Cos $\omega_0 t u(t)$]
 Neso Academy
 80K views · 6 years ago

**Laplace Transform
[Sin $\omega_0 t u(t)$]**

$\cos(\omega_0 t)u(t) \Rightarrow \frac{s}{s^2 + \omega_0^2}$; $\text{Re}\{s\} > 0$

$f(t) = \sin(\omega_0 t)u(t) \Rightarrow F(s) = ?$; ROC = ?

$f(t) = \frac{1}{2j} [e^{j\omega_0 t} - e^{-j\omega_0 t}] u(t)$

$F(s) = \frac{1}{2j} [e^{j\omega_0 t} u(t) - e^{-j\omega_0 t} u(t)]$

243 Signal & System **6:48**

Laplace Transform of Basic
 Signals [Sin $\omega_0 t u(t)$]
 Neso Academy
 61K views · 6 years ago

**Laplace Transform
(Solved Problem I)**

Question: Consider the function $f(t)$ having Laplace transform

$F(s) = \frac{\omega_0}{s^2 + \omega_0^2}$ **GATE 2006**

The final value of $f(t)$ would be

(a) 0 (b) 1
 (c) $-1 \leq f(\infty) \leq 1$ (d) ∞

244 Signal & System **7:02**

Laplace Transform (Solved
 Problem 1)
 Neso Academy
 91K views · 6 years ago



233 Signal & System **9:30**

Neso Academy
81K views · 6 years ago

Laplace Transform Properties PART 6

11. Differentiation in frequency:
 $f(t) \xrightarrow{\text{FT}} F(s), \text{Re } s = \sigma$
 $t^n f(t) \xrightarrow{\text{FT}} (-1)^n \frac{d^n F(s)}{ds^n}, \text{Re } s = \sigma$
 $F(s) \xrightarrow{\text{IFT}} f(t) = \int_{-\infty}^{\infty} F(s) e^{st} ds$
 $\frac{dF(s)}{ds} = \int_{-\infty}^{\infty} t f(t) e^{st} ds$

12. Integration in frequency:
 $f(t) \xrightarrow{\text{FT}} F(s)$
 $\frac{f(t)}{t} \xrightarrow{\text{FT}} \int_s^{\infty} F(u) du$
 $F(s) \xrightarrow{\text{IFT}} f(t) = \int_{-\infty}^{\infty} F(s) e^{st} ds$
 $\int_s^{\infty} F(u) du = \int_{-\infty}^{\infty} \int_s^{\infty} F(u) e^{st} ds du$

234 Signal & System

Properties of Laplace Transf...
Neso Academy
76K views · 6 years ago

Initial Value Theorem

13. Initial Value Theorem:
 $f(t) \xrightarrow{\text{FT}} F(s)$
 $f(0^+) = \lim_{s \rightarrow \infty} s F(s)$

Conditions: 1. Applicable only when $f(t) = 0, t < 0$
 2. $f(t)$ must not contain any impulse or higher order singularities at $t = 0$

235 Signal & System **12:47**

Initial Value Theorem
Neso Academy
1.3 lakh views · 6 years ago

Final Value Theorem

14. Final Value Theorem:
 $f(t) \xrightarrow{\text{FT}} F(s)$
 $\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} s F(s)$

Conditions: 1. Applicable only when $f(t) = 0, t < 0$
 2. $sF(s)$ must have poles in the left half of the complex plane

236 Signal & System **7:13**

Final Value Theorem
Neso Academy
1.1 lakh views · 6 years ago

Laplace Transform (Unit Impulse Signal)

$f(t) = \delta(t) \xrightarrow{\text{FT}} F(s)$
 $F(s) = \int_{-\infty}^{\infty} \delta(t) e^{-st} dt = e^{-s \cdot 0} = 1$
 $F(s) = 1$

237 Signal & System **4:55**

Laplace Transform of Basic Signals (Unit Impulse Signal)
Neso Academy
1 lakh views · 6 years ago

Laplace Transform (Unit Step Signal)

$f(t) = u(t) \xrightarrow{\text{FT}} F(s)$
 $F(s) = \int_{-\infty}^{\infty} u(t) e^{-st} dt = \int_0^{\infty} 1 \cdot e^{-st} dt = \left[-\frac{1}{s} e^{-st} \right]_0^{\infty} = \frac{1}{s}$

5:01

Laplace Transform of Basic Signals (Unit Step Signal)
Neso Academy

Laplace Transform of Basic



Laplace Transform Properties **PART 5**

10. Integration in Time:

$$f(t) \xleftrightarrow{\quad} F(s), \operatorname{Re} s = \sigma$$

$$\int_{-\infty}^{\infty} f(t) dt \xleftrightarrow{\quad} \frac{F(s)}{s}, \operatorname{Re} s = \operatorname{Re} \sigma + 1$$

$$\int_{-\infty}^t f(t) dt \xleftrightarrow{\quad} \frac{F(s)}{s} + \frac{f(0^-)}{s}$$

233 Signal & System **9:30**

Properties of Laplace Transform (Part 5)

Neso Academy

81K views · 6 years ago

Laplace Transform Properties **PART 6**

11. Differentiation in Frequency:

$$t^n f(t) \xleftrightarrow{\quad} (-1)^n \frac{d^n F(s)}{ds^n}, \operatorname{Re} s = \sigma$$

12. Integration in Frequency:

$$\frac{f(t)}{t} \xleftrightarrow{\quad} \int_s^{\infty} F(u) du$$

$$\int_{-\infty}^{\infty} f(t) dt = F(0)$$

234 Signal & System **8:50**

Properties of Laplace Transform (Part 6)

Neso Academy

76K views · 6 years ago

Initial Value Theorem

13. Initial Value Theorem:

$$f(t) \xleftrightarrow{\quad} F(s)$$

$$f(0^+) = \lim_{s \rightarrow \infty} s F(s)$$

Conditions:

1. Applicable only when $f(t) = 0, t < 0$
2. $f(t)$ must not contain any impulse or higher order singularities at $t = 0$

235 Signal & System

Initial Value Theorem

Neso Academy

1.3 lakh views · 6 years ago



Final Value Theorem

14. Final Value Theorem:

$$f(t) \xleftrightarrow{\quad} F(s)$$

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} s F(s)$$

Conditions:

1. Applicable only when $f(t) = 0, t < 0$
2. $sF(s)$ must have poles in the left half of the s-plane

236 Signal & System **7:13**

Final Value Theorem

Neso Academy

1.1 lakh views · 6 years ago

Laplace Transform (Unit Impulse Signal)

$$f(t) = \delta(t) \xleftrightarrow{\quad} F(s)$$

$$F(s) = \int_{-\infty}^{\infty} \delta(t) e^{-st} dt = 1$$

237 Signal & System **4:55**

Laplace Transform of Basic Signals (Unit Impulse Signal)

Neso Academy

1 lakh views · 6 years ago

Laplace Transform

Laplace Transform of Basic

**254** Signal & System**Invers Laplace Transform (Solved Problem)**Question: The Laplace Transform of a signal $y(t)$ is $Y(s) = 3/(s+3)$, then its final value is

- (a) -3 (b) 0
(c) 3 (d) Unbounded **GATE 2007**

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} s Y(s) = \lim_{s \rightarrow 0} s \cdot \frac{3}{s+3} = \frac{0}{3} = 0$$

8:52

255 Signal & System**Laplace Transform to Solve Differential Equations**

Question: A continuous time LTI system is described by

$$\frac{d^2 y(t)}{dt^2} + 4 \frac{dy(t)}{dt} + 5y(t) = 2 \frac{dx(t)}{dt} + 4x(t)$$

The response $y(t)$ of the system for the input $x(t) = e^{-2t} u(t)$ is _____

13:07

257 Signal & System**Laplace Transform (Solved problem 12)**

Question: The differential equation of an LTI system is given below

$$\frac{d^2 y(t)}{dt^2} - \frac{dy(t)}{dt} - 2y(t) = x(t)$$

Determine $h(t)$ in each of the following cases

- i) When the system is stable
ii) When the system is causal
iii) When the system is neither causal nor stable

14:47

258 Signal & System**Laplace Transform (Solved problem 13)**Question: For the given differential equation, find $y(t)$ under zero input condition.

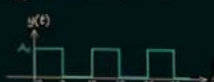
$$\frac{d^2 y(t)}{dt^2} + 6 \frac{dy(t)}{dt} + 8y(t) = x(t)$$

Initial conditions: $y(0^-) = 1$ and $y'(0^-) = 0$

9:49

259 Signal & System**Laplace Transform (Solved problem 14)**Question: $f(t) = F(s)$

$$y(t) = \sum_{n=0}^{\infty} f(t - nT_s) = Y(s) = ?$$



10:35

260 Signal & System**Inverse Laplace Transform (Solved Problem)**

Neso Academy

47K views · 6 years ago

Laplace Transform to Solve Differential Equations

Neso Academy

73K views · 6 years ago

Laplace Transform (Solved Problem 12)

Neso Academy

44K views · 6 years ago

Laplace Transform (Solved Problem 13)

Neso Academy

39K views · 6 years ago

Laplace Transform (Solved Problem 14)

Neso Academy

34K views · 6 years ago



Initial Value Theorem

13. Initial Value Theorem:

$$f(t) \longleftrightarrow F(s)$$

$$f(0^+) = \lim_{s \rightarrow \infty} s F(s)$$

Conditions: 1. Applicable only when $f(t) = 0, t < 0$
2. $f(t)$ must not contain any impulse or higher order singularities at $t = 0$

12:47

235/Signal & System

Initial Value Theorem

Neso Academy

1.3 lakh views · 6 years ago



Final Value Theorem

14. Final Value Theorem:

$$f(t) \longleftrightarrow F(s)$$

$$\lim_{t \rightarrow \infty} f(t) = f(\infty) = \lim_{s \rightarrow 0} s F(s)$$

Conditions: 1. Applicable only when $f(t) = 0, t < 0$
2. $sF(s)$ must have poles in the left half plane

7:13

236/Signal & System

Final Value Theorem

Neso Academy

1.1 lakh views · 6 years ago



Laplace Transform (Unit Impulse Signal)

$$f(t) = \delta(t) \longleftrightarrow F(s)$$

$$F(s) = \int_{-\infty}^{\infty} \delta(t) e^{-st} dt = \int_{-\infty}^{\infty} \delta(t) dt = 1$$

$$F(s) = 1$$



237/Signal & System

Laplace Transform of Basic ...

Neso Academy

1 lakh views · 6 years ago



Laplace Transform (Unit Step Signal)

$$f(t) = u(t) \longleftrightarrow F(s) = ?$$

$$F(s) = \int_{-\infty}^{\infty} u(t) e^{-st} dt = \int_0^{\infty} 1 \cdot e^{-st} dt = \left[-\frac{e^{-st}}{s} \right]_0^{\infty} = \frac{1}{s}$$

5:01

238/Signal & System

Laplace Transform of Basic Signals (Unit Step Signal)

Neso Academy

81K views · 6 years ago



Laplace Transform (DC Value & Signum Function)

$$1) f(t) = A_s \longleftrightarrow F(s) = ?$$

$$F(s) = \int_{-\infty}^{\infty} A_s e^{-st} dt = A_s \left[-\frac{e^{-st}}{s} \right]_{-\infty}^{\infty} = \frac{A_s}{s}$$

$$2) f(t) = \text{sgn}(t) = \frac{u(t) - u(-t)}{2}$$

$$F(s) = \frac{1}{2} \left[\frac{1}{s} - \frac{1}{-s} \right] = \frac{1}{s^2}$$

12:18

239/Signal & System

Laplace Transform of Basic Signals (DC Value & Signum...

Neso Academy

59K views · 6 years ago



Laplace Transform

Laplace Transform of Basic Signals



Laplace Transform of Basic Signals (Unit Step Signal)

$$f(t) = u(t) \Rightarrow F(s) = ?$$

$$\begin{aligned} F(s) &= \int_{-\infty}^{+\infty} u(t) \cdot e^{-st} dt \\ &= \int_{-\infty}^0 0 \cdot e^{-st} dt + \int_0^{\infty} 1 \cdot e^{-st} dt \\ &= 0 + \left[\frac{e^{-st}}{-s} \right]_0^{\infty} \quad (1) \end{aligned}$$

$$F(s) = -\frac{1}{s} \left[\underset{\downarrow 0}{e^{-s \cdot \infty}} - \underset{\downarrow 1}{e^{-s \cdot 0}} \right]$$

$$F(s) = -\frac{1}{s}(-1) \Rightarrow F(s) = \frac{1}{s} \Rightarrow \boxed{u(t) \Rightarrow \frac{1}{s}} \quad \text{ROC: } \text{Re}\{s\} > 0$$



$$\begin{aligned} u(t) &= 1 \cdot u(t) \\ &= e^{0t} \cdot u(t) \\ \sigma > 0 &\Rightarrow \text{Re}\{s\} > 0 \end{aligned}$$

Laplace Transform | Chapt... · 16/37

Neso Academy



Final Value Theorem

14. Final Value Theorem:

$$f(t) \Rightarrow F(s)$$

$$\lim_{t \rightarrow \infty} f(t) = f(\infty) = \lim_{s \rightarrow 0} s F(s)$$

Conditions: 1. Applicable only when $f(t) = 0, t < 0$
2. $sF(s)$ must have poles in the left half of the complex plane.

7:13

236 Signal & System

Final Value Theorem

Neso Academy

1.1 lakh views · 6 years ago

Laplace Transform (Unit Impulse Signal)

$$f(t) = \delta(t) \Rightarrow F(s) = ?$$

$$F(s) = \int_{-\infty}^{\infty} \delta(t) e^{-st} dt$$

$$F(s) = \int_{-\infty}^{\infty} \delta(t) e^{-st} dt$$

$$F(s) = 1$$

4:55

237 Signal & System

Laplace Transform of Basic Signals (Unit Impulse Signal)

Neso Academy

1 lakh views · 6 years ago

Laplace Transform (Unit Step Signal)

$$f(t) = u(t) \Rightarrow F(s) = ?$$

$$F(s) = \int_{-\infty}^{\infty} u(t) e^{-st} dt$$

$$F(s) = \int_{-\infty}^{\infty} u(t) e^{-st} dt$$

$$F(s) = \frac{1}{s}$$



238 Signal & System

Laplace Transform of Basic ...

Neso Academy

81K views · 6 years ago



Laplace Transform (DC Value & Signum Function)

$$1) f(t) = A_0 \Rightarrow F(s) = ?$$

$$F(s) = \int_{-\infty}^{\infty} A_0 e^{-st} dt$$

$$F(s) = \int_{-\infty}^{\infty} A_0 e^{-st} dt$$

$$F(s) = \frac{A_0}{s}$$

$$2) f(t) = \text{sgn}(t) \Rightarrow F(s) = ?$$

$$F(s) = \int_{-\infty}^{\infty} \text{sgn}(t) e^{-st} dt$$

$$F(s) = \int_{-\infty}^{\infty} \text{sgn}(t) e^{-st} dt$$

$$F(s) = \frac{1}{s^2}$$

12:18

239 Signal & System

Laplace Transform of Basic Signals (DC Value & Signum...)

Neso Academy

59K views · 6 years ago

Laplace Transform (Exponential Signals)

$$1) f(t) = e^{-at} u(t) \Rightarrow F(s) = ?$$

$$F(s) = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-st} dt$$

$$F(s) = \int_{-\infty}^{\infty} e^{-at} u(t) e^{-st} dt$$

$$F(s) = \frac{1}{s+a}$$

$$2) f(t) = -e^{-at} u(-t) \Rightarrow F(s) = ?$$

$$F(s) = \int_{-\infty}^{\infty} -e^{-at} u(-t) e^{-st} dt$$

$$F(s) = \int_{-\infty}^{\infty} -e^{-at} u(-t) e^{-st} dt$$

$$F(s) = \frac{1}{s+a}$$

9:51

240 Signal & System

Laplace Transform of Basic Signals (Exponential Signals)

Neso Academy

1.3 lakh views · 6 years ago

Laplace Transform

Laplace Transform of Basic

Laplace Transform of Basic Signals (Exponential Signals)

1) $f(t) = e^{-at} u(t) \Rightarrow 1/s+a$; $\text{Roc} : \text{Re}\{s\} > -a$

2) $f(t) = -e^{-at} u(-t)$

3) $f(t) = e^{at} u(t)$

4) $f(t) = -e^{at} u(-t)$

$\rightarrow f(t) = e^{-at} u(t) \Rightarrow F(s) = ?$

$$F(s) = \int_{-\infty}^{\infty} e^{-at} u(t) \cdot e^{-st} dt$$

$$= 0 + \int_0^{\infty} e^{-at} e^{-st} dt$$

$$= \int_0^{\infty} e^{-(s+a)t} dt$$

$$F(s) = \left[\frac{e^{-(s+a)t}}{-(s+a)} \right]_0^{\infty}$$

$$F(s) = \frac{-1(0-1)}{s+a} = \frac{1}{s+a}$$

$$= -\frac{1}{s+a} \left[\underbrace{e^{-(s+a)\infty}}_0 - \underbrace{e^0}_1 \right]$$

$$\sigma + a > 0$$

$$\Rightarrow \boxed{\sigma > -a} \text{ Roc}$$

$$s = \sigma + j\omega$$

$$e^{-(\sigma + j\omega + a)\infty}$$

$$= \frac{-1}{s+a} = \frac{-1}{\sigma + j\omega + a}$$

$$\underbrace{e^{-(\sigma+a)\infty}}_{e^{-\infty} = 0} \cdot \underbrace{e^{-j\omega\infty}}_{\text{finite (und.)}}$$

$$e^{-\infty} = 0 \quad \sqrt{\cos \infty - j \sin \infty}$$

Laplace Transform of Basic Signals (Exponential Signals)

1) $f(t) = e^{-at} u(t) \Rightarrow 1/s+a ; \text{ROC} : \text{Re}\{s\} > -a$

2) $f(t) = -e^{-at} u(-t) \Rightarrow 1/s+a ; \text{ROC} : \text{Re}\{s\} < -a$

3) $f(t) = e^{at} u(t) \Rightarrow ?$

4) $f(t) = -e^{at} u(-t) \Rightarrow 1/s-a ; \text{ROC} : \text{Re}\{s\} < a$

→ $f(t) = \underbrace{e^{-at} u(t)}_{\text{RSS}} \Rightarrow F(s) = ?$

$$F(s) = \int_{-\infty}^{\infty} e^{-at} u(t) \cdot e^{-st} dt$$

$$= 0 + \int_0^{\infty} e^{-at} e^{-st} dt$$

$$= \int_0^{\infty} e^{-(s+a)t} dt$$

$$F(s) = \left[\frac{e^{-(s+a)t}}{-(s+a)} \right]_0^{\infty}$$

$$= -\frac{1}{s+a} \left[\underbrace{e^{-(s+a)\infty}}_0 - \underbrace{e^0}_1 \right]$$

$$F(s) = -\frac{1(0-1)}{s+a} = \frac{1}{s+a}$$

$$\sigma + a > 0$$

$$\Rightarrow \boxed{\sigma > -a} \text{ ROC}$$

$$\boxed{\sigma > -a}$$

$$s = \sigma + j\omega$$

$$e^{-(\sigma + j\omega + a)\infty}$$

$$= \frac{1}{s+a}$$

$$\underbrace{e^{-(\sigma+a)\infty}}_{e^{-\infty}} \cdot \underbrace{e^{-j\omega\infty}}_{\text{finite (und.)}}$$

$$e^{-\infty} = 0 \quad \sqrt{\cos \infty - j \sin \infty}$$

Laplace Transform of Basic Signals (Exponential Signals)

1) $f(t) = e^{-at} u(t) \Rightarrow 1/s+a ; \text{ROC} : \text{Re}\{s\} > -a$

2) $f(t) = -e^{-at} u(-t) \Rightarrow 1/s+a ; \text{ROC} : \text{Re}\{s\} < -a$

3) $f(t) = e^{at} u(t) \Rightarrow ?$

4) $f(t) = -e^{at} u(-t) \Rightarrow 1/s-a ; \text{ROC} : \text{Re}\{s\} < a$

→ $f(t) = \underbrace{e^{-at} u(t)}_{\text{RSS}} \Rightarrow F(s) = ?$

$$F(s) = \int_{-\infty}^{\infty} e^{-at} u(t) \cdot e^{-st} dt$$

$$= 0 + \int_0^{\infty} e^{-at} e^{-st} dt$$

$$= \int_0^{\infty} e^{-(s+a)t} dt$$

$$F(s) = \left[\frac{e^{-(s+a)t}}{-(s+a)} \right]_0^{\infty}$$

$$= -\frac{1}{s+a} \left[\underbrace{e^{-(s+a)\infty}}_0 - \underbrace{e^0}_1 \right]$$

$$F(s) = -\frac{1(0-1)}{s+a} = \frac{1}{s+a}$$

$$\sigma + a > 0$$

$$\Rightarrow \boxed{\sigma > -a} \text{ ROC}$$

$$\boxed{\sigma > -a}$$

$$s = \sigma + j\omega$$

$$e^{-(\sigma+j\omega+a)\infty}$$

$$= \frac{1}{s+a} = \frac{1}{\sigma + j\omega + a}$$

$$e^{-(\sigma+a)\infty} \cdot e^{-j\omega\infty}$$

$$e^{-\infty} = 0 \quad \text{finite (und.)}$$

$$e^{-\infty} = 0 \quad \cos \infty - j \sin \infty$$